

High-Amplitude and Prolonged Glucose Excursions as a Key Determinant of Discordance Between Glucose Management Indicator and Glycated Hemoglobin in Type 1 Diabetes

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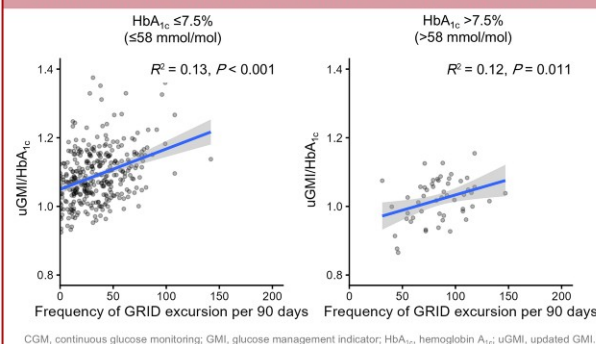
Objective

- How do specific glucose excursion patterns captured by CGM contribute to GMI-HbA_{1c} discordance in type 1 diabetes?

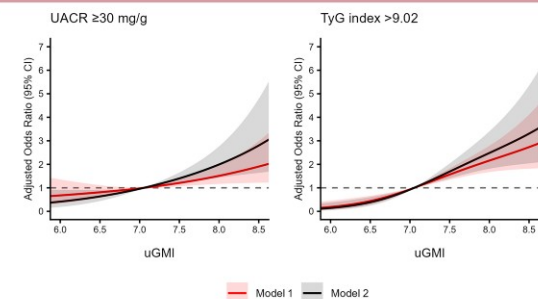
Method

- 90-day CGM data from 611 individuals were paired with HbA_{1c} results obtained within ± 15 days. Glucose excursions were assessed using the glucose rate increase detector algorithm with varied peak glucose and time-to-peak thresholds.

High-amplitude, prolonged glucose excursions as a consistent determinant of GMI-HbA_{1c} discordance across HbA_{1c} subgroups



Effect of GRID-derived excursion adjustment on glycemic risk gradients for UACR ≥ 30 mg/g and TyG index > 9.02



Model 1 was adjusted for age, sex, BMI, hypertension, and eGFR, with additional adjustment for RAS inhibitor and SGLT2 inhibitor use (UACR ≥ 30 mg/g) or dyslipidemia (TyG index > 9.02).

Model 2 additionally included GRID excursion frequency.

eGFR, estimated glomerular filtration rate; GRID, glucose rate increase detector; RAS, renin-angiotensin system; SGLT2, sodium-glucose transporter 2; TyG, triglyceride-glucose; UACR, urine albumin-to-creatinine ratio; uGMI, updated GMI.

Conclusion

- High-amplitude and prolonged glucose excursions were consistently associated with GMI-HbA_{1c} discordance and selectively strengthened uGMI-based risk gradients for albuminuria and elevated TyG index.

ARTICLE HIGHLIGHTS

Why did we undertake this study?

Glucose management indicator–hemoglobin A_{1c} (GMI-HbA_{1c}) discordance is common in diabetes, yet its physiological basis and clinical implications remain unclear.

What is the specific question(s) we wanted to answer?

We examined whether continuous glucose monitoring (CGM)-detected glucose excursion patterns explain GMI-HbA_{1c} discordance in adults with type 1 diabetes.

What did we find?

High and prolonged glucose excursions identified by the glucose rate increase detector algorithm were consistently associated with GMI-HbA_{1c} discordance across CGM platforms, HbA_{1c} strata, and analytic conditions. Accounting for excursion structure selectively enhanced the clinical informativeness of GMI and updated GMI for albuminuria and elevated triglyceride-glucose index compared with HbA_{1c}.

What are the implications of our findings?

Glucose excursion structure represents a physiological contributor to GMI-HbA_{1c} discordance and refines the interpretation of CGM-derived indices beyond HbA_{1c}.



High-Amplitude and Prolonged Glucose Excursions as a Key Determinant of Discordance Between Glucose Management Indicator and Glycated Hemoglobin in Type 1 Diabetes

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OBJECTIVE

Discordance between the glucose management indicator (GMI) and hemoglobin A_{1c} (HbA_{1c}) is frequently observed in diabetes, yet its physiological basis remains unclear. This study investigated how specific glucose excursion patterns captured by continuous glucose monitoring (CGM) contribute to this discordance in individuals with type 1 diabetes.

RESEARCH DESIGN AND METHODS

Ninety-day CGM traces from 611 adults with type 1 diabetes were paired with HbA_{1c} results obtained within ± 15 days. Glucose excursions were quantified using the glucose rate increase detector (GRID) algorithm with varied peak glucose and time-to-peak thresholds. Discordance was defined using GMI-to-HbA_{1c} and updated GMI (uGMI)-to-HbA_{1c} ratios, and associations with GRID-derived excursion metrics were evaluated alongside conventional CGM-derived variability metrics.

RESULTS

Excursions with peak glucose ≥ 250 mg/dL and time to peak ≥ 90 min were significantly associated with higher uGMI-to-HbA_{1c} ratios after adjustment for age, sex, estimated glomerular filtration rate, and HbA_{1c} group, with consistent findings across CGM devices (sensor type 1: $\beta = 0.174$, 95% CI 0.147–0.201; sensor type 2: $\beta = 0.102$, 95% CI 0.068–0.136; both $P < 0.001$) and alternative GMI formulations. In restricted cubic spline analyses, adjustment for GRID-derived excursion metrics differentially reshaped the associations of HbA_{1c}, GMI, and uGMI with albuminuria and elevated triglyceride-glucose index in an outcome- and context-dependent manner, preferentially enhancing the informativeness of GMI and uGMI but not HbA_{1c}.

CONCLUSIONS

Frequent high and prolonged glucose excursions were consistently associated with GMI-HbA_{1c} discordance across devices, HbA_{1c} strata, and analytic conditions. GRID-derived excursion metrics modify the relationship between GMI/uGMI and glycemia-associated risk.



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Hemoglobin A_{1c} (HbA_{1c}) has long been the cornerstone for assessing long-term glycemic management and predicting the risk of microvascular and macrovascular complications in diabetes (1,2). With the widespread adoption of continuous glucose monitoring (CGM), the glucose management indicator (GMI)—an estimate of HbA_{1c} derived from mean CGM glucose—has become increasingly used to complement laboratory HbA_{1c} in clinical practice (3). However, discordance between GMI and laboratory-measured HbA_{1c} is frequently observed, complicating interpretation of CGM data (4).

Accumulating evidence suggests that this discordance is not merely a technical artifact (4–6). Prior studies have linked GMI-HbA_{1c} discordance to hematologic and renal factors that influence erythrocyte lifespan and glycation efficiency (7,8), and to higher risks of albuminuria and cardiovascular events, indicating potential clinical relevance beyond average glucose exposure alone (9–11). Efforts to address regression-related bias led to the development of an updated GMI (uGMI) equation, which modestly improved alignment with HbA_{1c}, particularly at lower glucose levels, yet substantial discordance persists (12).

Most prior investigations incorporating CGM metrics relied on summary measures of glycemic variability, such as the coefficient of variation (CV), which quantify overall variability but overlook the temporal dynamics of glucose excursions—specifically, their amplitude and duration—that may disproportionately influence both mean glucose and its short-term representation by GMI (13,14).

Advances in algorithmic glucose pattern analysis now allow for automated detection of glucose excursions using raw CGM data. In particular, the glucose rate increase detector (GRID) enables standardized identification of excursions characterized by sharp and prolonged glucose rises following meals, offering a more physiologically relevant approach to understanding GMI-HbA_{1c} discordance.

We hypothesized that individuals with pronounced glucose excursions may exhibit higher GMI relative to HbA_{1c}, since GMI reflects recent mean glucose levels and responds promptly to transient hyperglycemia, whereas HbA_{1c} reflects cumulative glycemic exposure over a longer time frame.

Using synchronized 90-day CGM and laboratory HbA_{1c} data from adults with

type 1 diabetes, we examined which excursion characteristics were most strongly associated with GMI-HbA_{1c} discordance and whether accounting for excursion structure modified the relationships between HbA_{1c}, GMI, uGMI, and early metabolic risk markers, including albuminuria and the triglyceride-glucose (TyG) index—an indicator of glucose-lipid interactions related to insulin resistance and increased cardiovascular risk (15,16).

RESEARCH DESIGN AND METHODS

Data Source and Study Population

We studied adults aged ≥ 18 years with type 1 diabetes using 90-day raw CGM data collected at the endocrinology outpatient clinic of Samsung Medical Center, Republic of Korea, between May 2019 and January 2025. The diagnosis of type 1 diabetes was based on the reimbursement criteria of the Korean National Health Insurance Service, which require insulin use and at least one of the following: fasting C-peptide ≤ 0.2 nmol/L, glucagon- or meal-stimulated C-peptide ≤ 0.6 nmol/L, 24-h urine C-peptide < 30 $\mu\text{g/day}$, positivity for glutamic acid decarboxylase or other islet autoantibodies, or a history of diabetic ketoacidosis at diagnosis (17–19). CGM records were retrieved from a cloud-based repository, and clinical data were obtained from the Clinical Data Warehouse of Samsung Medical Center. The index date was defined as the final day of the 90-day monitoring period, and the nearest HbA_{1c} measurement within ± 15 days of this date was used. Fasting plasma glucose, urine albumin-to-creatinine ratio (UACR), and triglycerides (TG) were collected within ± 30 days, and other laboratory variables were collected within ± 60 days.

Initially, 1,011 individuals who had used either sensor type 1 (S1) (Dexcom G5, G6, or G7; Dexcom, San Diego, CA) or sensor type 2 (S2) (FreeStyle Libre [FSL] 1 or 2; Abbott Diabetes Care, Wiltney, Oxon, U.K.) for at least 90 days, and for whom HbA_{1c}, UACR, fasting glucose, TG, and CGM metrics (with sensor wear time $\geq 70\%$) were available, were screened at Samsung Medical Center. Of these, participants were excluded if they had an estimated glomerular filtration rate (eGFR) < 30 mL/min per 1.73 m² within the past 6 months ($n = 27$), anemia defined as hemoglobin < 10 g/dL within the

past 6 months ($n = 15$), pregnancy within the previous 1 year ($n = 9$), a prior diagnosis of cancer ($n = 96$), a history of oral or intravenous steroid use within the past 6 months ($n = 89$), or type 2 diabetes ($n = 164$). A total of 611 individuals were included in the final analysis (Supplementary Fig. 1).

This study was approved by the Institutional Review Board of Samsung Medical Center, Seoul, Republic of Korea, (SMC no. 2025-11-054), which waived the requirement for written informed consent. All procedures were conducted in accordance with the Declaration of Helsinki.

Clinical Variables and CGM Metrics

Clinical variables included age, sex, BMI, eGFR, HbA_{1c}, fasting glucose, fasting C-peptide, liver enzymes (AST and ALT), lipid parameters (total cholesterol, TG, HDL cholesterol, and LDL cholesterol), and medication use (insulin, metformin, sodium-glucose cotransporter 2 [SGLT2] inhibitors, glucagon-like peptide 1 receptor agonists, statins, fibrates, n-3 fatty acids, and renin-angiotensin system inhibitors). Albuminuria was defined as UACR ≥ 30 mg/g. Hypertension was defined as systolic blood pressure ≥ 140 mmHg, diastolic blood pressure ≥ 90 mmHg, use of antihypertensive medications, or one or more claims with ICD-10 codes I10–I11. Dyslipidemia was defined as total cholesterol ≥ 6.22 mmol/L or one or more claims with ICD-10 code E78 plus at least one prescription of a lipid-lowering agent per year. Chronic kidney disease was defined as eGFR < 60 mL/min per 1.73 m².

CGM metrics included time in range (TIR), time in tight range (TITR), time above range (TAR), time below range (TBR), mean glucose, mean amplitude of glycemic excursion (MAGE), continuous overall net glycemic action (CONGA), CV, GMI, uGMI (12), and sensor wear time. For MAGE and CONGA, CGM data were preferentially analyzed in three 30-day windows with $\geq 70\%$ data coverage; otherwise, the full CGM period was used. Discordance was quantified using GMI-to-HbA_{1c} and uGMI-to-HbA_{1c} ratios.

Detection and Characterization of Glucose Excursions Using GRID

Glucose excursions were identified from raw CGM time series data using a modified implementation of GRID (20). Excursion

onset required glucose ≥ 130 mg/dL with a sustained rate of increase. For each excursion, a validated peak glucose level and time to peak were determined using a sequential peak validation approach (Supplementary Fig. 2).

Excursion magnitude was classified according to validated peak glucose thresholds of ≥ 140 , ≥ 180 , and ≥ 250 mg/dL, based solely on the maximum glucose level reached during the excursion, without imposing any minimum duration above these thresholds. Time to peak was defined as the interval between excursion onset and the validated peak. For descriptive and inferential analyses, time to peak was categorized as < 60 min, 60 to < 90 min, or ≥ 90 min. In addition, prolonged excursions were operationally defined using a time-to-peak criterion of ≥ 90 min among excursions with a validated peak glucose ≥ 250 mg/dL. These GRID-derived metrics capture discrete features of excursion dynamics—including rise pattern, peak magnitude, and temporal persistence—that are conceptually distinct from cumulative duration-based measures such as TAR (> 180 and > 250 mg/dL).

CGM Devices and Data Processing

Only CGM data with $\geq 70\%$ sensor wear time were included. The sensor wear time $\geq 70\%$ CGM data criterion was defined as overall sensor wear time across the entire 90-day monitoring period, applied at the participant level; days with a high proportion of missing CGM values were not excluded separately. Data excluded implausible readings (e.g., > 400 or < 40 mg/dL unless sustained for ≥ 10 min) and applied smoothing filters for short-term noise. The excursion count, average peak value, and time-to-peak duration were derived per subject. GMI and uGMI (12) were calculated based on the full 90-day data set.

CGM data were obtained from S1 and S2 sensors, each providing glucose readings at 5-min and 15-min intervals, respectively. To harmonize detection sensitivity, a modified version of the GRID algorithm calibrated for 15-min sampling was applied to S2 data. For each device group, the number of GRID-detected excursions showed a linear correlation with the corresponding uGMI values, thereby confirming a near-proportional relationship

between excursion frequency and mean glucose exposure (Supplementary Fig. 3).

Under the assumption that uGMI reflects comparable mean glucose exposure across devices, excursion frequencies observed in the S2 group were referenced to S1-equivalent values based on this linear scaling relationship. These S1-equivalent values were used only as an approximation to illustrate the linear correspondence between raw S2 and S1 excursion counts, rather than as adjusted or substituted measurements.

Equivalence Testing Between CGM Systems

To evaluate the analytical comparability of GMI-HbA_{1c} relationships across CGM systems, we conducted two one-sided equivalence tests (21,22). Comparisons were performed between independent groups using different CGM systems (S1, G6 vs. G7; S2, FSL 1 vs. FSL 2) and between manufacturers (S1 vs. S2), based on the uGMI-to-HbA_{1c} ratio. No individual wore multiple devices simultaneously. Dexcom G5 sensors ($n = 4$) were excluded because of insufficient sample size. The equivalence margin was prespecified at ± 0.1 , corresponding to $\sim 10\%$ of the mean ratio value. This threshold was chosen a priori on the basis of the analytical accuracy of current CGM systems, whose mean absolute relative difference typically ranges from 8% to 10% when compared with reference plasma glucose measurements (23). Given that the GMI is derived directly from CGM glucose data, a $\pm 10\%$ variation in the GMI-to-HbA_{1c} ratio was considered to represent the inherent measurement variability of the device rather than a clinically meaningful difference. Equivalence was concluded when both one-sided tests rejected the null hypothesis that the true mean difference exceeded the equivalence bounds ($P < 0.05$ for both sides).

Statistical Analysis

Continuous variables are presented as mean (SD) or median (interquartile range), and categorical variables are presented as number (%). Baseline clinical characteristics and CGM metrics of participants were compared between the two CGM groups using Student *t* test or Mann-Whitney *U* test for continuous variables and Pearson χ^2 test for categorical variables. Additionally, participants within each CGM group

were further categorized by HbA_{1c} levels ($< 6.5\%$ [48 mmol/mol], 6.5–7.5% [48–58 mmol/mol], and $> 7.5\%$ [58 mmol/mol]) for stratified analyses.

We conducted a supplementary analysis to assess whether the linear association between GRID excursion frequency and uGMI was consistent across CGM devices with different glucose data intervals (5 min in the S1 group and 15 min in the S2 group). Scatter plots with fitted linear regression lines were generated separately for the S1 and S2 groups, restricting the analysis to GRID excursions with a time to peak longer than 90 min, at glucose peak thresholds (≥ 140 , ≥ 180 , and ≥ 250 mg/dL). Linear regression was performed to evaluate the association.

Multivariable linear regression assessed associations between GRID excursion frequency and discordance ratios, adjusting for age, sex, eGFR, and HbA_{1c} strata, with analyses stratified by CGM device. The total 90-day frequency of GRID excursions (defined as a peak glucose ≥ 250 mg/dL and time to peak ≥ 90 min) was divided by 90 to represent the frequency per day.

Associations of HbA_{1c}, GMI, and uGMI with albuminuria (defined as UACR ≥ 30 mg/g) and high TyG index (> 9.02) (15) were evaluated using restricted cubic spline analyses, with complementary receiver operating characteristic (ROC) analysis assessing relative discrimination. As a supplementary stratified analysis, participants were categorized into tertiles of GRID excursion frequency based on cohort-specific cut points. Within each tertile, ROC curves and areas under the curve (AUC) were calculated for HbA_{1c}, GMI, and uGMI to discriminate a high TyG index (> 9.02). DeLong test was used to compare the AUCs between HbA_{1c} and uGMI.

The adjusted odds ratios for albuminuria and high TyG index based on HbA_{1c}, GMI, and uGMI as continuous variables were visualized using restricted cubic spline analyses with three knots. For each variable (HbA_{1c}, GMI, and uGMI), the median value was set as the reference point. Both models were adjusted for age, sex, BMI, hypertension, and eGFR. The model for albuminuria was additionally adjusted for RAS inhibitor and SGLT2 inhibitor use, while the model for high TyG index was additionally adjusted for dyslipidemia.

As sensitivity analyses, we examined the robustness of the association between GRID excursion frequency and

the uGMI-to-HbA_{1c} ratio across CGM devices and HbA_{1c} subgroups by 1) restricting the analysis to CGM data obtained over a 30-day period and 2) applying an Asian-specific, population-derived GMI formulation (24), using the same analytic framework as in the primary analyses.

In exploratory analyses, participants with HbA_{1c} >7.5% (58 mmol/mol) were first stratified according to CGM-derived glycemic variability metrics, defined as values exceeding the metric-specific median. Within each device- and metric-specific stratum, participants were further classified into low and high uGMI-HbA_{1c} groups using a median split of the uGMI-HbA_{1c} group. Twenty-four-hour glucose profiles were compared using raw CGM data aggregated over 90 days. Random forest models were fitted as a sensitivity analysis to benchmark discrimination against logistic regression using pre-specified predictor sets (HbA_{1c}, uGMI, uGMI + GRID excursions, and HbA_{1c} + uGMI + GRID excursions) for UACR ≥30 mg/g and TyG index >9.02. Performance was evaluated by fivefold cross-validated AUC (mean ± SD), with additional models using balanced class weights to address class imbalance.

All analyses were performed using R software (version 4.4.3; R Foundation for Statistical Computing). Statistical significance was defined as a two-sided $P < 0.05$.

Data and Resource Availability

The data sets generated and/or analyzed during the current study are not publicly available because of institutional and patient privacy restrictions but are available from the corresponding author upon reasonable request and with approval from the institutional review board.

RESULTS

Clinical Characteristics of Participants

Among the 611 participants, the mean age was 45.8 ± 16.6 years, and 42.6% were male. Baseline characteristics and CGM metrics are summarized in Table 1. Using a predefined equivalence margin of ±0.1 for the uGMI-to-HbA_{1c} ratio, intra-device comparisons demonstrated equivalence within S1 (G6 vs. G7, $P < 0.001$) and within S2 (FSL 1 vs. FSL 2, $P < 0.001$). Cross-manufacturer comparison between pooled S1 and S2 users

also met the equivalence criterion ($P < 0.001$), supporting analytical comparability across systems (Supplementary Fig. 4).

Association Between GRID Excursion Frequency and GMI-HbA_{1c} Discordance

Stronger associations between GRID excursion frequency and uGMI-HbA_{1c} were observed with increasing peak glucose thresholds and longer time-to-peak categories across both CGM platforms (Fig. 1 and Supplementary Fig. 5), with attenuation at lower thresholds (Supplementary Figs. 5 and 6). These associations were consistent across HbA_{1c} strata (<6.5% [48 mmol/mol], 6.5–7.5% [48–58 mmol/mol], and >7.5% [58 mmol/mol]). The uGMI-to-HbA_{1c} ratio showed numerically greater associations with GRID excursion frequency than the GMI-to-HbA_{1c} ratio, particularly for excursions with peak ≥250 mg/dL and time to peak ≥90 min (Supplementary Fig. 7). The 30-day sensitivity analysis yielded results directionally consistent with the primary 90-day analysis. In selected subgroups—particularly among S2 users with HbA_{1c} 6.5–7.5% (48–58 mmol/mol) and >7.5% (58 mmol/mol)—the magnitude of association was numerically greater than in the 90-day analysis (Supplementary Table 1 and Supplementary Fig. 8). Sensitivity analyses using an Asian-specific GMI formulation (24) yielded concordant results (Supplementary Figs. 9 and 10).

In multivariable linear regression stratified by CGM system, GRID excursion (peak ≥250 mg/dL and time to peak ≥90 min) frequency remained independently associated with uGMI-HbA_{1c} and GMI-HbA_{1c} after adjustment for age, sex, eGFR, and HbA_{1c} strata (S1, $\beta = 0.174$ [95% CI 0.147–0.201] and 0.116 [95% CI 0.092–0.140]; S2, $\beta = 0.102$ [95% CI 0.068–0.136] and 0.068 [95% CI 0.039–0.097]; all $P < 0.001$).

Association Between Conventional Glycemic Variability Metrics and GMI-HbA_{1c} Discordance

In contrast, CV—a mean-independent metric of glycemic variability—showed weak and inconsistent associations with GMI-to-HbA_{1c} and uGMI-to-HbA_{1c} discordance ratios, with substantially lower explanatory power than GRID excursion frequency. This pattern was most evident among participants with HbA_{1c} >7.5% (58 mmol/mol), in whom associations

were attenuated or directionally inconsistent (Fig. 2A). By comparison, mean-dependent glycemic variability and exposure metrics—including MAGE, CONGA, and TAR-based indices—showed robust and consistent positive associations with GMI-HbA_{1c} and uGMI-HbA_{1c} discordance in participants with HbA_{1c} ≤7.5% (58 mmol/mol). In contrast, among participants with HbA_{1c} >7.5% (58 mmol/mol), these associations diverged by CGM platform (Fig. 2B–D and Supplementary Fig. 11). Within this subgroup, S1 users were characterized by extremely low T1TR and markedly elevated mean glucose levels (Supplementary Table 2). Notably, among participants exceeding the median values of mean-dependent variability and TAR-based metrics, those with lower uGMI-to-HbA_{1c} ratios in the S1 group exhibited prolonged hyperglycemia throughout most of the day, whereas this pattern was not observed in the corresponding S2 group (Supplementary Fig. 12). Associations between CV and mean-dependent glycemic variability metrics across HbA_{1c} strata are shown in Supplementary Fig. 13.

Risk Gradients and Discrimination of HbA_{1c}, GMI, and uGMI for Albuminuria and TyG Index

Restricted cubic spline analyses were used to characterize how HbA_{1c}, GMI, and uGMI relate to albuminuria (UACR ≥30 mg/g) and a high TyG index (>9.02) across the glycemic spectrum, whereas ROC analyses assessed their relative discriminative performance (Fig. 3). In spline analyses, all three markers showed nonlinear risk gradients; however, adjustment for GRID-derived excursion metrics selectively altered the shape of these gradients for GMI and uGMI, with comparatively little effect on HbA_{1c}. For albuminuria, excursion adjustment consistently reshaped the risk gradients of GMI and uGMI across HbA_{1c} levels. For the TyG index, similar modifications were observed predominantly at higher HbA_{1c} levels (>7.0% [53 mmol/mol]), with minimal changes at lower HbA_{1c} levels.

In ROC analyses, unadjusted models demonstrated comparable discrimination among HbA_{1c}, GMI, and uGMI for both outcomes. After adjustment for baseline clinical covariates (model 1), overall discrimination improved. Additional adjustment for GRID-derived excursion metrics produced outcome- and context-dependent changes in discrimination (model 2). For

Table 1—Baseline characteristics and CGM metrics of study participants

	Overall	S1 group	S2 group
<i>n</i>	611	390	221
Age, years	45.75 (16.57)	42.56 (14.80)	51.39 (18.00)
Male sex, <i>n</i> (%)	260 (42.60)	150 (38.50)	110 (49.80)
BMI, kg/m ² *	24.04 (3.71)	24.02 (3.73)	24.07 (3.70)
Diabetes duration, years	14.14 (10.08)	13.96 (9.98)	14.45 (10.27)
Dyslipidemia, <i>n</i> (%)	494 (80.90)	315 (80.80)	179 (81.00)
Hypertension, <i>n</i> (%)	224 (36.70)	131 (33.60)	93 (42.10)
SBP, mmHg†	125.72 (16.13)	124.31 (15.58)	128.61 (16.89)
DBP, mmHg†	71.91 (12.27)	72.29 (12.01)	71.14 (12.78)
Use of RAS inhibitors, <i>n</i> (%)	108 (17.70)	60 (15.40)	48 (21.70)
Use of metformin, <i>n</i> (%)	78 (12.80)	33 (8.50)	45 (20.40)
Use of GLP-1RA, <i>n</i> (%)	16 (2.60)	13 (3.30)	3 (1.40)
Use of SGLT2 inhibitors, <i>n</i> (%)	29 (4.70)	16 (4.10)	13 (5.90)
Use of statins, <i>n</i> (%)	347 (56.80)	218 (55.90)	129 (58.40)
Use of fibrates, <i>n</i> (%)	3 (0.50)	2 (0.50)	1 (0.50)
Use of n-3 fatty acids, <i>n</i> (%)	9 (1.50)	5 (1.30)	4 (1.80)
UACR, mg/g, median (IQR)	7.52 (4.30–18.30)	6.88 (4.07–16.60)	8.38 (4.68–25.00)
UACR group, <i>n</i> (%)			
UACR <30 mg/g	500 (81.90)	327 (83.80)	173 (78.30)
UACR ≥30 mg/g	111 (18.10)	63 (16.20)	38 (21.70)
HbA _{1c} , %	6.83 (0.94)	6.69 (0.92)	7.09 (0.91)
HbA _{1c} , mmol/mol	51 (10.00)	49.61 (10.06)	53.97 (9.96)
HbA _{1c} group, <i>n</i> (%)			
<6.5% (48 mmol/mol)	224 (36.70)	172 (44.10)	52 (23.50)
6.5–7.5% (48–58 mmol/mol)	272 (44.50)	166 (42.60)	106 (48.00)
>7.5% (58 mmol/mol)	115 (18.80)	52 (13.30)	63 (28.50)
Hemoglobin, g/dL‡	13.72 (1.73)	13.69 (1.74)	13.78 (1.72)
Fasting C-peptide, ng/mL§	0.30 (0.41)	0.23 (0.39)	0.42 (0.41)
AST, units/L	24.24 (9.79)	23.95 (9.33)	24.74 (10.56)
ALT, units/L	20.48 (12.36)	19.95 (12.14)	21.42 (12.70)
Total cholesterol, mg/dL	163.17 (35.36)	165.10 (34.57)	159.76 (36.55)
Triglycerides, mg/dL	82.77 (56.96)	79.19 (53.97)	89.09 (61.51)
HDL cholesterol, mg/dL	68.37 (17.76)	70.24 (17.42)	65.08 (17.92)
LDL cholesterol, mg/dL	78.23 (28.60)	79.01 (28.28)	76.84 (29.17)
eGFR, mL/min/1.73 m ²	98.52 (20.33)	100.91 (19.63)	94.32 (20.91)
Fasting glucose, mg/dL	146.62 (51.86)	141.93 (49.33)	154.90 (55.20)
TyG index	8.52 (0.63)	8.45 (0.63)	8.65 (0.60)
CGM metrics			
Sensor wear time, %	91.93 (8.01)	92.98 (7.99)	90.07 (7.73)
Mean glucose, mg/dL	160.54 (29.74)	160.69 (28.54)	160.29 (31.81)
CV, %	35.54 (5.62)	35.65 (5.51)	35.36 (5.81)
TIR (70–180 mg/dL [3.9–10.0 mmol/L]), %	65.96 (16.54)	65.88 (16.19)	66.10 (17.16)
TITR (70–140 mg/dL [3.9–7.8 mmol/L]), %	42.83 (15.78)	42.63 (15.21)	43.19 (16.76)
TAR >140 mg/dL (7.8 mmol/L), %	54.43 (16.62)	54.67 (15.99)	54.01 (17.69)
TAR >180 mg/dL (10.0 mmol/L), %	31.31 (17.06)	31.42 (16.68)	31.10 (17.75)
TAR >250 mg/dL (13.9 mmol/L), %	10.08 (10.95)	10.12 (10.64)	10.02 (11.51)
TBR <54 mg/dL (3.0 mmol/L), %	0.46 (0.68)	0.56 (0.75)	0.28 (0.51)
TBR <70 mg/dL (3.9 mmol/L), %	2.73 (2.47)	2.70 (2.34)	2.80 (2.68)

Continued on p. 1226

Table 1—Continued

	Overall	S1 group	S2 group
GMI	7.15 (0.71)	7.15 (0.68)	7.14 (0.76)
uGMI	7.18 (0.89)	7.19 (0.85)	7.17 (0.94)
GMI-to-HbA _{1c} ratio	1.05 (0.08)	1.08 (0.08)	1.01 (0.07)
uGMI-to-HbA _{1c} ratio	1.06 (0.08)	1.08 (0.08)	1.01 (0.07)

Values are presented as mean (SD), median (IQR), or number (%). DBP, diastolic blood pressure; GLP-1RA, glucagon-like peptide 1 receptor agonist; IQR, interquartile range; SBP, systolic blood pressure. *Missing data (n = 4). †Missing data (n = 161). ‡Missing data (n = 426). §Missing data (n = 307).

albuminuria, GMI and uGMI showed higher AUCs than HbA_{1c} when evaluated in the full cohort (DeLong test $P = 0.014$). For the TyG index, discrimination did not differ significantly overall; however, stratification by excursion burden revealed higher discrimination for GMI and uGMI than HbA_{1c} in the highest tertile of GRID-derived excursions (Supplementary Fig. 14).

To distinguish their potential role as independent predictors from their modifier effects, we further evaluated models in which GRID-derived excursion metrics were directly added alongside uGMI. In these multivariable logistic regression models, discrimination did not improve meaningfully. Similarly, in exploratory random forest analyses, inclusion of GRID-derived metrics resulted in numerically higher AUCs in crude models, but these differences were not statistically significant (DeLong test; $P = 0.509$ for UACR, $P = 0.053$ for TyG index) and were attenuated after covariate adjustment (Supplementary Fig. 15).

CONCLUSIONS

In this study, high-amplitude and prolonged glucose excursions automatically detected using the GRID algorithm were a consistent and independent determinant of discordance between GMI/uGMI and HbA_{1c} across CGM devices, HbA_{1c} strata, and statistical models. In spline analyses, adjustment for GRID-derived excursion metrics differentially reshaped the associations of HbA_{1c}, GMI, and uGMI with albuminuria and elevated TyG index in an outcome- and context-dependent manner, with a preferential impact on GMI and uGMI.

HbA_{1c} reflects cumulative glycation over the erythrocyte lifespan (4,5,7,25) and is thus relatively insensitive to short-term glycemic fluctuations (3). In contrast, GMI (and uGMI) is derived directly from mean CGM glucose and responds immediately to transient hyperglycemia. Our findings suggest that, when glucose

excursions are both high in amplitude and prolonged in duration, GMI rises disproportionately relative to HbA_{1c}, resulting in measurable discordance. Importantly, the observed discordance was not merely statistical: higher GMI/uGMI aligned with complication-related phenotypes captured by albuminuria and elevated TyG index with performance that was at least comparable to that of HbA_{1c}.

Prior investigations of GMI-HbA_{1c} discordance have emphasized demographic, hematologic, or renal determinants of glycation efficiency, and CGM-based studies have often relied on global summary measures of variability such as CV (14). In our analyses, CV showed weak and inconsistent associations with GMI-to-HbA_{1c} and uGMI-to-HbA_{1c} ratios, reinforcing that mean-independent dispersion does not reliably capture the glycemic dynamics that drive discordance. By contrast, multiple mean-dependent metrics (e.g., GRID excursion frequency, TAR, MAGE, and CONGA) were positively and, in HbA_{1c} $\leq 7.5\%$ (58 mmol/mol) strata, robustly associated with GMI-HbA_{1c} and uGMI-HbA_{1c} discordance. Although these associations diverged at higher HbA_{1c} levels, their overall pattern supports that discordance reflects excursion burden and sustained hyperglycemia rather than variability independent of mean glucose.

Notably, the attenuation and heterogeneity observed in participants with HbA_{1c} $> 7.5\%$ (58 mmol/mol) were phenotype-specific. Among S2 users, mean-dependent metrics retained explanatory value, whereas a subset of S1 users with markedly low T1TR and prolonged hyperglycemia exhibited reduced discrimination by TAR-, MAGE-, and CONGA-based measures—consistent with partial saturation of duration- or magnitude-based indices under persistently high glucose exposure. These observations may help explain why prior studies did not consistently identify such associations, particularly when analyses were not stratified by

HbA_{1c} range or glycemic phenotype. In contrast, GRID-derived excursion metrics retained explanatory value even in this context, suggesting that discrete excursion structure captures complementary dimensions of glycemia beyond cumulative exposure alone.

The spline and ROC analyses together indicate that incorporating excursion structure can meaningfully alter how glycemic markers relate to complication-related outcomes. For albuminuria, adjustment for GRID-derived excursion metrics more prominently reshaped the risk gradients for GMI and uGMI than for HbA_{1c} across the glycemic spectrum, and discrimination for albuminuria favored GMI/uGMI over HbA_{1c} in the overall cohort. For the TyG index, the effect of excursion adjustment on risk gradients for GMI/uGMI was more evident at higher HbA_{1c} levels, and discrimination improved preferentially in individuals with substantial excursion burden (i.e., the highest tertile of GRID excursions). This pattern suggests that excursion structure informs the interpretation of CGM-derived indices in an outcome- and context-dependent manner rather than uniformly improving all markers.

Complementary modeling further clarified the role of GRID-derived excursion metrics. When excursion metrics were used as adjustment variables (i.e., to account for excursion structure) in spline and ROC analyses, the apparent informativeness of GMI and uGMI for albuminuria and, under high excursion burden, TyG index was selectively strengthened. However, when excursion metrics were added as independent predictors alongside uGMI in predictive models, overall performance did not consistently improve: logistic regression showed no meaningful gain for either outcome, and random forest models showed, at most, numerically higher AUCs in crude settings that were attenuated after covariate adjustment. This apparent discrepancy is

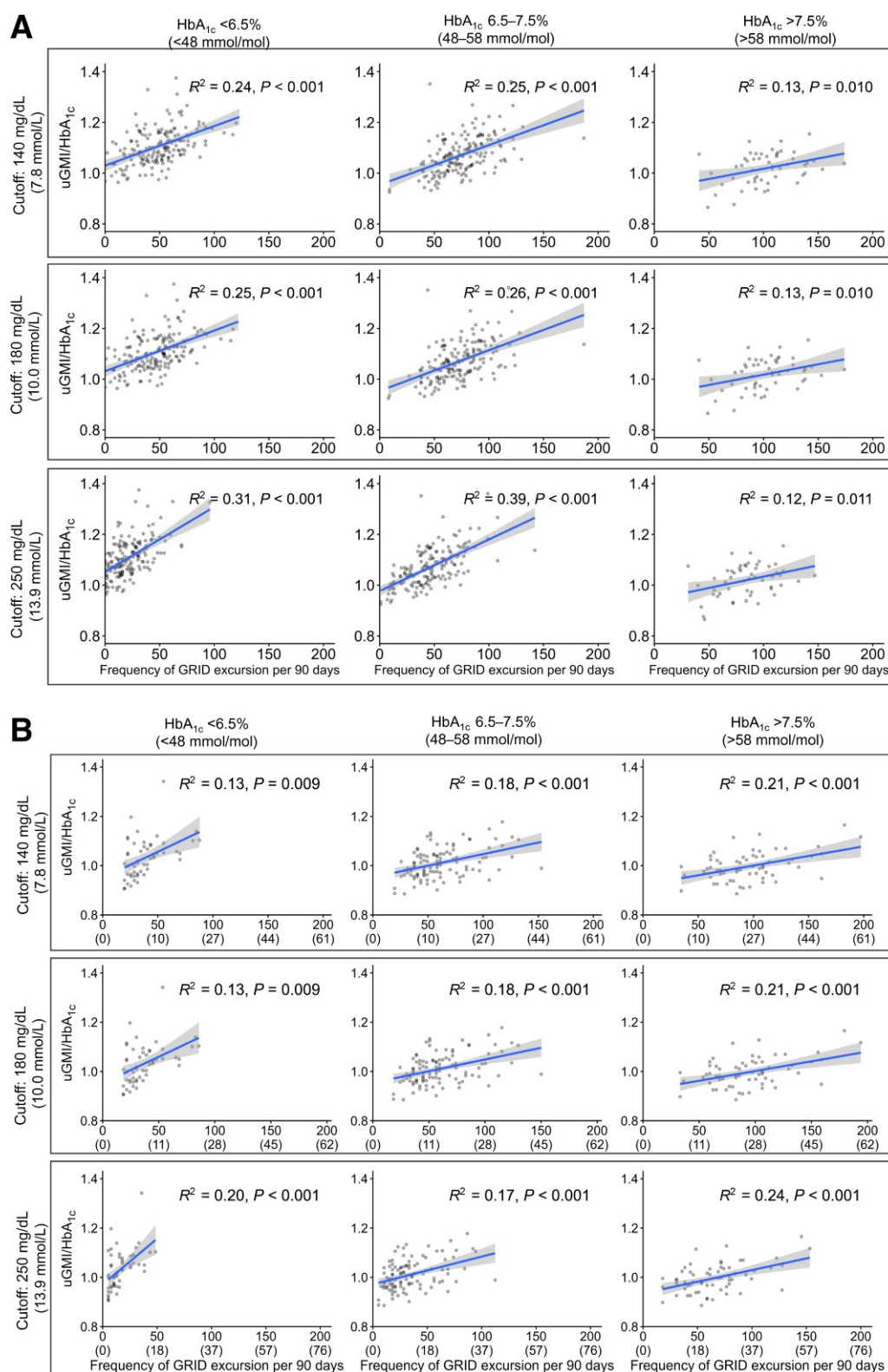


Figure 1—Relationship between GRID excursion frequency and uGMI-to-HbA_{1c} ratio across CGM sensor types and HbA_{1c} subgroups (time to peak ≥ 90 min). Participants were stratified by baseline HbA_{1c} level (columns: <6.5% [48 mmol/mol], 6.5–7.5% [48–58 mmol/mol], and >7.5% [58 mmol/mol]). Rows correspond to excursion magnitude categories defined by validated peak glucose thresholds (≥ 140 , ≥ 180 , and ≥ 250 mg/dL [7.8, 10.0, and 13.9 mmol/L]). Each point represents an individual participant and reflects the frequency of GRID-detected excursions exceeding the corresponding peak glucose threshold, with excursion magnitude defined based solely on the maximum glucose level reached during each excursion. A: S1 group. B: S2 group. In B, x-axis tick labels represent S1-equivalent values, with the corresponding raw S2 values shown in parentheses. Solid lines indicate linear regression fits with 95% CIs (shaded areas). Coefficients of determination (R^2) and P values are displayed in each panel.

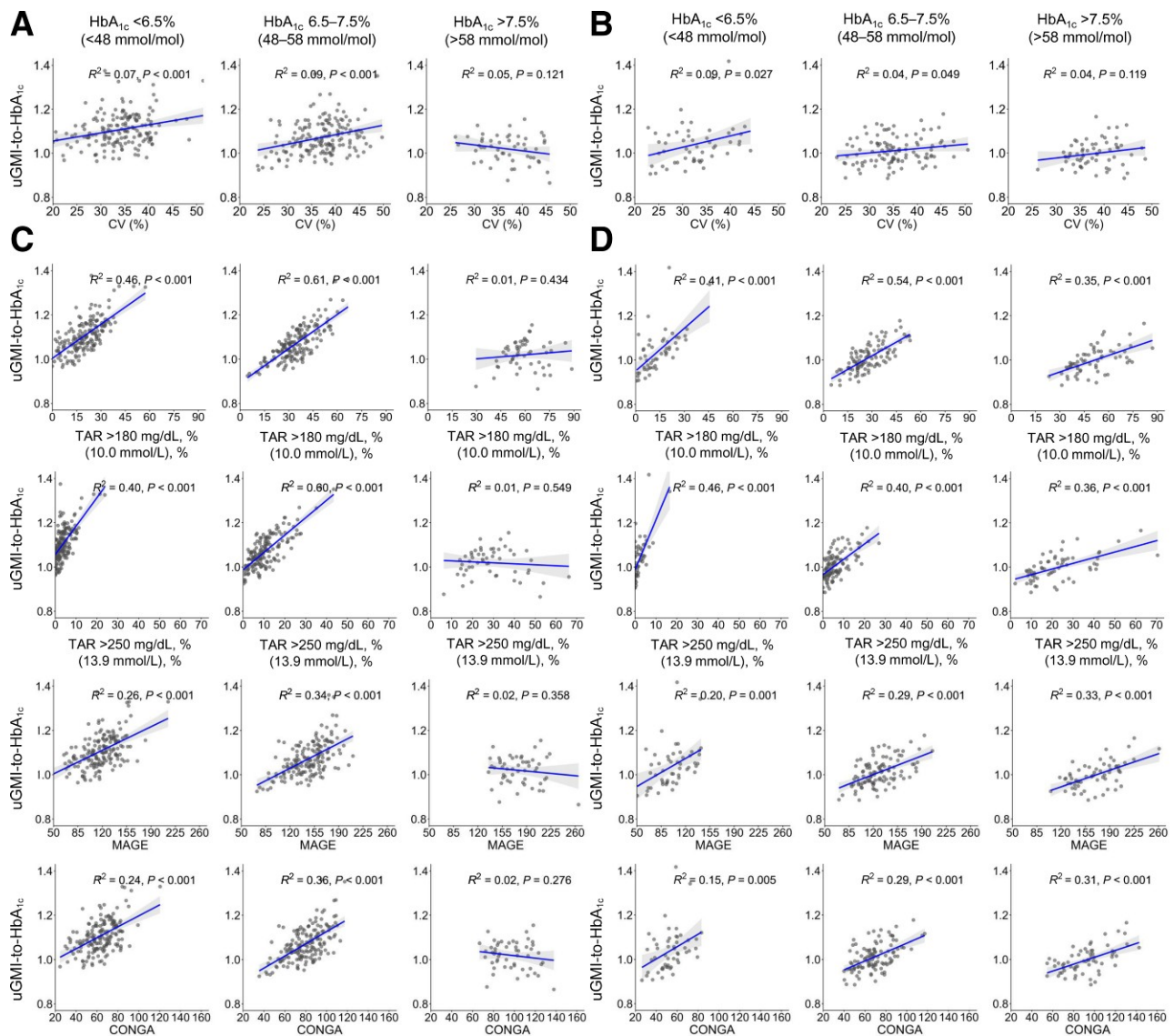


Figure 2—Associations between glycemic variability metrics and uGMI-to-HbA_{1c} ratio across CGM platforms and HbA_{1c} strata. A and B: Associations between CV (%) and the uGMI-to-HbA_{1c} ratio. C and D: Associations between mean-dependent variability metrics (TAR, MAGE, and CONGA) and the uGMI-to-HbA_{1c} ratio. A and C show the S1 group, and B and D show the S2 group. Participants were stratified by baseline HbA_{1c} level (columns: <6.5% [48 mmol/mol], 6.5–7.5% [48–58 mmol/mol], and >7.5% [58 mmol/mol]). Each point represents one participant. Solid lines indicate linear regression fits with 95% CIs (shaded areas). Coefficients of determination (R²) and P values are displayed in each panel.

consistent with a modifier role: excursion metrics may refine the relationship between CGM-derived indices and risk-related phenotypes—thereby changing the interpretation of GMI/uGMI—without necessarily contributing additive, standalone predictive information beyond established glycemic and clinical variables.

The present findings align with emerging evidence that CGM-derived metrics capture dimensions of glycemic physiology not fully reflected by HbA_{1c}. Large real-world studies have demonstrated associations between CGM-derived measures—including mean glucose, TIR, TAR, and glycemic variability—and vascular outcomes or

mortality independent of HbA_{1c} (26,27). In this context, our results provide mechanistic insight by demonstrating that CGM-derived indices encode temporal glycemic patterns that are averaged out in HbA_{1c} but may be relevant to early metabolic stress and complication risk.

This study extends prior literature in several ways. First, by leveraging raw 90-day CGM traces and automated excursion detection, it moves beyond summary variability measures to quantify excursion structure directly. Second, we evaluated both GMI and an updated regression formulation (uGMI). Although uGMI modestly improved alignment with HbA_{1c},

residual discordance remained common and was strongly linked to excursion structure, suggesting that refinement of the regression equation alone cannot fully reconcile CGM-derived indices and HbA_{1c} without considering temporal glycemic patterns. Third, our integrated analytic framework (linear models, spline analyses, ROC analyses, and sensitivity modeling) supports a coherent interpretation in which excursion structure modifies the clinical meaning of CGM-derived indices across glycemic contexts. The numerically stronger associations in the 30-day analyses may reflect tighter temporal coupling between recent CGM exposure and HbA_{1c}.

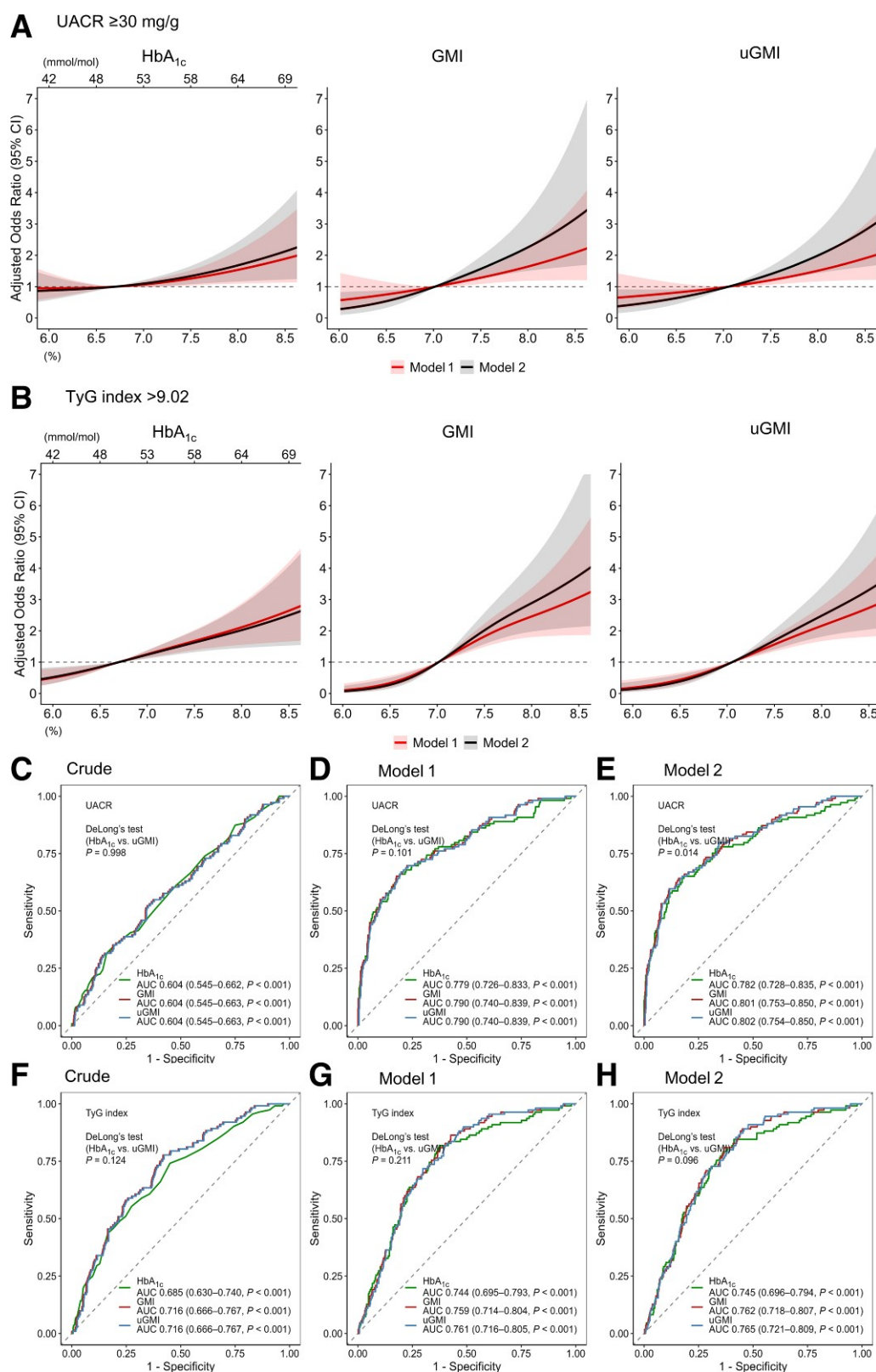


Figure 3—Restricted cubic spline and ROC analyses for predicting albuminuria (UACR ≥ 30 mg/g) and high TyG index (> 9.02). **A**: Adjusted model for UACR ≥ 30 mg/g. **B**: Adjusted model for TyG index > 9.02 . Model 1 was adjusted for age, sex, BMI (missing $n = 4$), hypertension, and eGFR, with additional adjustment for RAS inhibitor and SGLT2 inhibitor use (UACR ≥ 30 mg/g) or dyslipidemia (TyG index > 9.02). Model 2 additionally included GRID excursion frequency. **C–H** show ROC curves comparing the discriminative performance of HbA_{1c}, GMI, and uGMI for identifying UACR ≥ 30 mg/g (**C–E**) and TyG index > 9.02 (**F–H**). **C** and **F** show unadjusted models. **D** and **G** correspond to model 1, and **E** and **H** correspond to model 2, with the same covariate adjustments as described in **A** and **B**. Given the small number of participants taking fibrates or n-3 fatty acids, dyslipidemia was used instead of medication use for adjustment.

Several limitations should be acknowledged. The cross-sectional design precludes causal inference, particularly given prior evidence suggesting that the relationship between GMI and HbA_{1c} may vary across populations and may not remain stable within individuals over time (28). The cohort comprised adults with type 1 diabetes from a single tertiary center, which may limit generalizability. Although individuals with anemia or advanced kidney disease were excluded, residual contributors to HbA_{1c}-CGM discordance related to erythrocyte turnover cannot be fully excluded. In addition, S2 data were available for analysis at 15-min intervals as provided by the manufacturer, requiring a modified GRID implementation and approximate scaling to S1-equivalent excursion counts; these should be interpreted as illustrative rather than exact equivalents.

In conclusion, frequent GRID-detected high and prolonged glucose excursions were consistently associated with discordantly elevated GMI and uGMI relative to HbA_{1c} across CGM platforms, GMI formulations, HbA_{1c} strata, and analytic conditions. Accounting for excursion structure selectively strengthened the clinical informativeness of CGM-derived mean glucose indices in relation to albuminuria and, under high excursion burden, the TyG index—supporting a modifier role of GRID-derived excursion metrics that refines interpretation of CGM-derived indices rather than acting as an independent predictor.

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